Lumbopelvic Control and the Development of Upper Extremity Injury in Professional Baseball Pitchers

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Background: The baseball-throwing motion requires a sequential order of motions and forces initiating in the lower limbs and transferring through the trunk and ultimately to the upper extremity. Any disruption in this sequence can increase the forces placed on subsequent segments. No research has examined if baseball pitchers with less lumbopelvic control are more likely to develop upper extremity injury than pitchers with more control.

Purpose: To determine if baseball pitchers who sustain a chronic upper extremity injury have less lumbopelvic control before their injury compared with a group of pitchers who do not sustain an injury.

Study Design: Cohort study; Level of evidence, 2.

Methods: A total of 49 asymptomatic, professional baseball pitchers from a single Major League Baseball organization participated. Lumbopelvic control was measured using an iPod-based digital level secured to a Velcro belt around each player's waist to measure anteroposterior (AP) and mediolateral (ML) deviations (degrees) during single-leg balance with movement and static bridge maneuvers. During a competitive season, 22 of these pitchers developed upper extremity injuries, while the remaining 27 sustained no injuries. Separate 2-tailed *t*-tests were run to determine if there were significant differences in lumbopelvic control between groups ($P < .05$).

Results: There were no significant between-group differences for the stride leg (nondominant) during the bridge test in either the AP ($P = .79$) or the ML ($P = .42$) directions, or either direction during the drive leg bridge test ($P > .68$). However, the injured group had significantly less lumbopelvic control than the noninjured group during stride leg balance in both the AP ($P = .03$) and the ML ($P = .001$) directions and for drive leg balance in both the AP ($P = .01$) and the ML ($P = .04$) directions.

Conclusion: These results demonstrate that baseball pitchers with diminished lumbopelvic control, particularly during stride leg and drive leg single-leg balance with movement, had more upper extremity injuries than those with more control. Clinicians should consider evaluating lumbopelvic control in injury prevention protocols and provide appropriate exercises for restoring lumbopelvic control before returning athletes to competition after injury. Specific attention should be given to testing and exercises that mimic a single-limb balance task.

Keywords: baseball; biomechanics; elbow; motion analysis; pelvis; shoulder

Overuse upper extremity injuries are a common occurrence among baseball players, with pitchers being the most susceptible.[9,15,21] This is not surprising considering professional pitchers often throw in excess of 85 to 100 pitches per game[18,23] and create shoulder and elbow torques exceeding 103 N·m and 96 N·m, respectively.[8] In a recent study conducted among Major League Baseball players, shoulder injuries comprised 35% of all upper extremity injuries, while elbow injuries were the second most with 31%.[9] Even more concerning was that these upper extremity injuries resulted in an average of 71 days on the injured list.

Proper transmission of forces through the kinetic chain during baseball pitching is believed to be critical for both performance and to reduce the risk of upper extremity injury.[2,6,22] Efficient force transmission allows for increased

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force production of the more proximal segments (ie, hips, core), while minimizing force on the more distal segments (ie, shoulder, elbow, wrist). Part of this kinetic chain involves significant lumbopelvic control (eg, muscle activation, dynamic control) to transmit forces developed in the lower extremity through the trunk to the upper extremity.[16,20] Baseball pitchers have been shown to have less lumbopelvic control during a single-leg bridge maneuver when compared with position players.[17] Research has also shown that these lumbopelvic deficiencies are at least partially associated with spending more days on the injured list.[4] These findings have been further supported by an investigation that showed moderate relationships between decreased lumbopelvic control with increased horizontal shoulder torque and elbow valgus torque while throwing a fastball.[16] Despite this negative relationship between lumbopelvic control and upper extremity forces during the pitching motion and the association with increased days on the injured list, there has been no research investigating if injury is more likely to develop in pitchers with less lumbopelvic control.

The purpose of this study was to investigate if differences exist in the bilateral lumbopelvic control between professional pitchers who develop upper extremity injury during a competitive season and those who do not develop injury. Keeley et al[13] reported that 78% of the peak glenohumeral compressive force during the throwing motion is associated with stride length, lateral pelvic tilt at maximum shoulder external rotation, and the velocity of axial pelvic rotation at ball release. On the basis of these findings, we hypothesized that pitchers who sustain a chronic upper extremity injury would have less lumbopelvic control before their injury, compared with a group of pitchers who do not sustain an injury. More specifically, the pitchers with less lumbopelvic control were hypothesized to have less control of their throwing-side leg during a single-leg bridge maneuver and less control of their non–throwing side leg during a single-leg dynamic balance test.

METHODS

Participants

Initially, 62 baseball pitchers, taken from a sample of convenience from a single professional baseball organization volunteered to participate in this study. Before data collection, all individuals had no injury within the past 6 months and none had any history of upper or lower extremity surgery.

After lumbopelvic control data collection, which occurred during spring training, participants were monitored for injuries by the organization's medical staff over the course of a competitive baseball season. The competitive season was defined as the first day of spring training until the end of each player's competitive season, which includes any playoff/championship games. Any participants who suffered a spine or lower extremity injury were not considered in the data analysis. Similarly, any participants who were released or traded by the organization were also not used for data analysis. This resulted in a total of 49 pitchers being used for the study. Of the 13 players excluded from the data analysis, 1 sustained a lumbar injury, 1 sustained a lower extremity injury, and the remaining players were either released or traded to another team. During the competitive season, 22 of these 49 pitchers developed upper extremity injuries, while the remaining 27 sustained no injuries at any point during the competitive season (Table 1). The individuals who sustained injury were evaluated and diagnosed by the organization's medical staff. These injuries were all diagnosed as chronic conditions and resulted in time lost from competition. The specific injuries consisted of 7 elbow ulnar collateral ligament sprains, 4 rotator cuff strains, 4 subacromial impingement/bursitis, 3 forearm strains, 1 elbow impingement, 1 biceps tendinitis, 1 ulnar neuropathy, and 1 elbow loose body.

TABLE 1
Participant Demographics[a]

	Injured Group (n = 22)	Noninjured Group (n = 27)	*P* Value
Age, y	25.1 ± 2.6	24.0 ± 2.8	.32
Height, cm	190.1 ± 6.0	190.7 ± 7.3	.99
Mass, kg	94.8 ± 10.0	94.2 ± 12.0	.72

[a]Data are reported as mean ± SD.

Procedures

This prospective cohort study examined if baseball pitchers who develop upper extremity injuries over the course of a competitive season have less lumbopelvic control at the beginning of spring training than those who do not sustain an injury. Lumbopelvic control testing occurred in a single session on the first day of spring training. All participants provided informed consent as mandated by the university's institutional review board before any data collection. All testing was completed by the same investigator (K.L.) and at the time of testing, no participants had any injuries.

Lumbopelvic control was measured using the Level Belt Pro (Perfect Practice Inc). The Level Belt Pro uses an iPod-based digital level secured to a Velcro belt (Figure 1) to determine the amount of anteroposterior (AP) and mediolateral (ML) deviations (degrees) from its starting position relative to a horizontal plane. Before data collection, the belt and attached iPod were positioned horizontally around each participant's waist covering the anterior and posterior superior iliac spines. Participants completed all testing in shorts, shirt, and no socks or shoes. For testing, each participant completed 4 tasks (bilateral single-leg balance with movement, bilateral static single-leg bridges). All tests were completed in a randomized order.

Single-Leg Balance With Movement Test. For the single-leg balance with movement test, participants initially stood with both feet resting on the ground. They were then instructed to shift their weight to the test leg, while flexing their non–test leg at the hip and knee, thereby elevating their test leg foot approximately 10 cm off the ground (Figure 2). This elevated position was held for 2

Figure 1. Level Belt Pro device for measuring lumbopelvic control.

Figure 2. Single-leg balance with movement test.

seconds and then the participant was instructed to slowly extended the non–test hip and knee returning the test leg back to the ground.[3] The investigator visually monitored each individual's technique throughout this movement to ensure correct motion was used and that participants did not hold onto any objects for support or allow their non–test limb to touch the ground. If at any time the correct motion was not performed, the test was terminated and repeated after 1 minute of rest. This test was then repeated on the opposite leg. A priori intratester reliability of this test was determined using an intraclass correlation coefficient [ICC(3, 2)] and standard error of measurement (SEM). The investigator of this study showed good reliability for the single-leg balance with movement test (ICC, 0.88; SEM, 0.93°).

Static Single-Leg Bridge Test. For the static single-leg bridge test, participants were positioned supine on a standard treatment table with their test knee and hip flexed and their foot resting on the table, while the non–test knee and hip were in a relaxed position.[17] They then performed a bridge maneuver by extending the test hip and lifting the pelvis off the table, so the test knee flexed to approximately 90°. At this same time, the non–test knee was fully extended, so the non–test leg aligned with the trunk (Figure 3). This position was held statically for 30 seconds. This test was then repeated on the opposite leg. A priori intratester reliability of this test was determined using an ICC(3, 2) and SEM. The investigator of this study showed good reliability for the static single-leg bridge test (ICC, 0.91; SEM, 0.61°).

Figure 3. Static single-leg bridge test.

Data Analysis

The absolute average amount of AP and ML lumbopelvic tilts measured during each test were used for data analyses. For lumbopelvic control testing, the "stride leg" was defined as the leg on the opposite side of the throwing arm. This leg is used to step or stride toward the home plate during the cocking and acceleration phases of the pitching motion. Conversely, the "drive leg" was defined as the leg on the same side as the throwing arm. This leg is used to develop power as the leg pushes off the pitching

TABLE 2
Between-Group Descriptive Statistics for Lumbopelvic Control[a]

Test	Injured	Noninjured	P Value	Effect Size
Single-Leg Balance With Movement				
Drive leg (AP)[b]	4.5 ± 3.2	2.6 ± 1.3	.01	1.5
Drive leg (ML)[b]	3.1 ± 1.7	2.3 ± 1.0	.04	0.8
Stride leg (AP)[b]	4.2 ± 2.2	2.8 ± 2.1	.03	0.7
Stride leg (ML)[b]	3.9 ± 2.1	2.2 ± 1.1	.001	1.6
Static Single-Leg Bridge				
Drive leg (AP)	2.2 ± 1.6	2.4 ± 1.7	.68	0.1
Drive leg (ML)	1.9 ± 1.6	1.9 ± 1.3	.71	0.0
Stride leg (AP)	2.3 ± 2.1	2.4 ± 2.1	.79	0.0
Stride leg (ML)	2.5 ± 1.6	2.2 ± 1.4	.42	0.2

[a]Data are reported as mean ± SD unless otherwise indicated. AP, anteroposterior; ML, mediolateral.
[b]Statistically significant difference between groups ($P < .05$).

mound rubber and directs the body forward toward the home plate.

Dependent variables included AP and ML lumbopelvic tilts for the single-leg balance with movement test and the static single-leg bridge test. Separate 2-tailed t-tests were calculated to determine the significant differences between the groups using SPSS Statistics Software (Version 26; IBM). All findings were considered significant at $P < .05$. To provide an indication of clinical significance, Cohen d between-group effect sizes for lumbopelvic control were calculated as (injured group – noninjured group) / noninjured group SD. Effect sizes were interpreted as small = 0.20, medium = 0.50, and large = 0.80.[5]

RESULTS

There were no significant differences for age, height, or mass between the injured and noninjured groups (Table 1). Descriptive between-group statistics for lumbopelvic control can be viewed in Table 2. There were no significant between-group differences for the stride leg (nondominant) during the bridge test in either the AP or the ML directions. Similarly, there were no significant differences for the drive leg (dominant) during the bridge test in either the AP or ML directions. However, the injured group had significantly less lumbopelvic control than the noninjured group during stride leg balance in both the AP and the ML directions and for drive leg balance in both the AP and the ML directions. These statistically significant differences also showed large effect sizes indicating clinical significance.

DISCUSSION

Pitching requires proper functioning of the kinetic chain to optimize performance and reduce stress on susceptible joints, such as those in the upper extremity. As such, clinicians have long speculated that deficiencies, such as limited lumbopelvic control, may increase a pitcher's risk of developing upper extremity injury. The results of this study are the first to show that pitchers with less lumbopelvic control at the beginning of spring training were more likely to develop an upper extremity injury than pitchers with more control.

Chaudhari et al[3,4] previously investigated the effect of lumbopelvic control on both pitching performance and its association with injury. In the first study, these investigators noted that professional baseball pitchers with less lumbopelvic control in their drive leg had poorer pitching performance (walks plus hits) than pitchers with better control.[3] They then conducted a subsequent study investigating lumbopelvic control and days on the injured list among professional pitchers.[4] This study reported that less AP control during the single-leg balance test in the drive leg was associated with spending more days on the injured list. Although, it is difficult to hypothesize how decreased control in either the ML or the AP direction directly affects the pitching motion, the combination of these motions may provide a more complete picture of a pitcher's overall lumbopelvic control. As such, good bilateral balance throughout the pitching motion may be important for proper transference of force and subsequently a reduced risk of injury.

Dillman et al[7] reported that the drive leg is the weight-bearing limb during the windup phase of the throwing motion, while the opposite limb moves into hip and knee flexion. These authors suggested that good balance during this phase of the pitching motion is critical for both performance and injury prevention. The injured pitchers in the current study demonstrated less control during a balance task that required a similar hip and knee motion to that of the windup, although without any trunk rotation. Proper balance on the drive leg has also been reported to affect stride length during pitching.[12] Similarly, lumbopelvic control of the stride leg is important for pitching as it contacts the ground during the cocking phase providing a stable base for proper kinematics and kinetics during the acceleration, deceleration, and follow-through phases of the throwing motion.[10,11,25] The arm cocking phase begins when the lead foot of the stride leg contacts the ground and functions as a stabilizer as the pelvis rotates toward the home plate.[10] Although good balance during the follow-through phase of pitching may not play a significant role in pitching performance, it may be critical in injury prevention as the hip and knee move into flexion and upper extremity forces are appropriately dissipated.[7] Based on these previous findings, it is conceivable that the decreased control present during the balance task in both the drive and the stride legs of the injured pitchers used in the current study may contribute to alterations during various phases of the pitching motion and contributed to their injury. However, future research is necessary to support this hypothesis.

As previously mentioned, the pitching motion creates a kinetic chain where motions and forces produced at 1 anatomic segment subsequently influence the motions and forces of the adjacent segment(s).[22] Kibler et al[14] speculated that the legs and core create the base upon which the entire pitch sequence is founded and produce most of the force that is ultimately imparted to the ball. However, if there is a lack of control of the lumbopelvic region, then force production may be regulated to the smaller and more injury-susceptible areas such as the shoulder, elbow, and forearm. Solomito et al[24] showed a significant relationship between trunk forward flexion at ball release and elbow valgus force, indicating a 2.9 N·m increase in valgus force for every 10° of trunk forward tilt. Oyama et al[19] reported that pitchers with excessive contralateral trunk tilt during the throwing motion produced unwarranted forces on the throwing shoulder and elbow, thereby increasing the pitchers' risk of injury. Similarly, Aguinaldo et al[1] reported that pitchers who began rotating their trunk before stride leg foot contact produced significantly more elbow valgus torque. Laudner et al[16] found that lumbopelvic control of the drive leg has a negative relationship with both shoulder and elbow torque during the pitching motion, meaning pitchers with less lumbopelvic control created more horizontal shoulder and elbow valgus torque. In a systematic review, Deal et al[6] noted that several lower extremity deficiencies, including decreased balance, were independent risk factors for elbow pain in throwing athletes. The results of the current study support these previous investigations, indicating that pitchers with less lumbopelvic control are more likely to develop upper extremity injury. This increased risk of injury may stem from the increased forces noted in these previous investigations.

There are several limitations of this study worth noting. The results of this study supported the authors' initial hypothesis that pitchers with less lumbopelvic control would be more likely to develop an upper extremity injury compared with pitchers with more control. However, these results do not discern what is the minimum level of lumbopelvic control necessary to decrease the risk of injury. Although lumbopelvic control has been identified as a key component for optimizing pitching performance and decreasing the risk of injury, it is just 1 segment of the kinetic chain. Discrepancies at any point in this chain may have detrimental effects on the distal joints and contribute to upper extremity injury. So, although diminished lumbopelvic control was identified in pitchers who developed injury, all segments of the kinetic chain should be addressed. Also, the methods used for assessing lumbopelvic control were chosen due to their reproducibility in a clinical setting, but they do not reflect the ballistic nature of the throwing motion. Although the authors of this study have shown strong intrarater reliability of the measurements used for assessing lumbopelvic control, there is no previous research assessing the validity of these measurements in relation to the kinematics of the throwing motion. This may partially explain why there were no significant between-group differences for either leg during the static single-leg bridge test. Upper extremity injuries are common in baseball.[9,15] However, the participants in this study were all professional pitchers and may have varying degrees of lumbopelvic control when compared with pitchers at other competitive levels. Future research is necessary to determine if differences exist in lumbopelvic control between competitive levels, as well as age levels and among athletes in other overhead sports.

CONCLUSION

The results of this study demonstrate that professional baseball pitchers with less lumbopelvic control may be more likely to develop an upper extremity injury compared with pitchers with more control. Clinicians should consider evaluating lumbopelvic control in injury prevention protocols and provide appropriate exercises for restoring lumbopelvic control before returning the athlete to competition after injury.

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